Haozengran Wang

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EDUCATION

Harbin Institute of Technology, School of Management **2023.09 – 2027.07**
Bachelor of Management in Big Data Management and Application | GPA: 85.3/100

emlyon business school **2023.09 – 2027.07**
Bachelor of Science in Applied Data Science | GPA: 85.3/100

RESEARCH INTERESTS

LLM Reasoning & Test-Time Scaling; Reliable Agentic Systems; Trustworthy LLMs & Alignment

PUBLICATIONS

AlphaMaster: Dual-Chain Feedback for Scalable and Diverse Alpha Factor Discovery
First Author | ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2026.05

From Predictive Signal to Deployable Value: EHR Measurement-Process Features in ICU AKI Prediction
Co-first Author (listed second) & Corresponding Author | International Journal of Medical Informatics, 221 (2026), 106668

PA-QPC: Reliability-Aware Queue-Pressure Signal Control Using Partial Connected Vehicle Data
Second Author & Corresponding Author | International Conference on Intelligent Transportation Engineering (ICITE 2026), Accepted 2026.07

NIPT Test Timing Recommendation via Statistical Learning & Intelligent Optimization
First Author | International Conference on Advanced Computation, Engineering Intelligence and Information Processing (ACEIIP 2026), 2025.12

MANUSCRIPTS UNDER REVIEW

Structural Factors Underlying Sufficient Reasoning Demand in Language Models
First Author | Manuscript under review at Communication AI and Computing

SAFE-MAS: Provenance-Aware Runtime Authorization for Tool-Augmented Multi-Agent LLM Systems
First Author | Manuscript under double-blind peer review at a major AI conference

Beyond Error Risk: Bounded Recoverability for Selective Control of Stateful Tool Actions
First Author | Manuscript under double-blind peer review at a major AI conference

Unwarranted Pragmatic Inference in LLM Writing Assistants: A Human-Calibrated Benchmark and Evaluation Audit
First Author | Manuscript under double-blind peer review at a major AI conference

PhysCrop-Risk: Calibrated Early Low-Yield Alerts from Phenology-Aligned Multispectral Satellite Time Series
First Author | Manuscript under double-blind peer review at a major AI conference

RESEARCH EXPERIENCE

Structural Factors Underlying Sufficient Reasoning Demand in Language Models
First Author | Manuscript under review at Communication AI and Computing

- Formulated sufficient reasoning demand as the minimum inference-time reasoning required to preserve near-optimal performance, and demonstrated across 530 problems from six benchmarks that demand is neither monotonically determined by task difficulty nor invariant across models.
- Constructed controlled task families isolating constraint density, search width, sequential depth, distractor load, and state-tracking load, showing that constraint density and search width shift sufficient demand while the remaining factors primarily increase reasoning expenditure.
- Developed and prospectively validated a frozen demand-aware allocation policy on 300 unseen problems, preserving pooled accuracy within a prespecified non-inferiority margin while reducing reasoning-token consumption by 31.9% versus an always-high baseline.

SAFE-MAS: Provenance-Aware Runtime Authorization for Tool-Augmented Multi-Agent LLM Systems
First Author | Manuscript under double-blind peer review

- Formulated tool-augmented multi-agent safety as a runtime authorization problem, addressing prompt injection, cross-agent trust laundering, and poisoned shared state, and distinguishing capability-compliant actions from actions grounded in user intent.
- Designed SAFE-MAS, a provenance-aware authorization framework integrating task–effect matching, cross-agent provenance, intent binding, structural state contracts, and selective confirmation for auditable Allow/Deny/Confirm decisions over tool calls and shared-state mutations.
- Evaluated SAFE-MAS across AgentDojo, an independent post-freeze audit, and shared-state attacks, observing no successful attacks across 432 injection trajectories, 1,080 audit trajectories, and 420 shared-state attacks, while reducing confirmations from 590 to 499 without an observed ASR increase.

Beyond Error Risk: Bounded Recoverability for Selective Control of Stateful Tool Actions
First Author | Manuscript under double-blind peer review

- Formalized bounded recoverability for stateful tool actions, capturing whether an erroneous action can be legally repaired within a finite action budget while restoring the user goal and satisfying policy constraints.
- Designed R²-CTRL, a recoverability-aware pre-commit controller that jointly models action-error risk and conditional recoverability to allocate Commit/Verify/Defer decisions, using offline counterfactual supervision with environment clones and depth-bounded recovery search.
- Evaluated R²-CTRL across held-out airline/retail tasks, long-horizon stress tests, matched-risk analysis, and distribution shifts, reducing bounded irrecoverable failure from 0.084 to 0.077 and read-only tool calls from 0.547 to 0.176 versus risk-only control.

Unwarranted Pragmatic Inference in LLM Writing Assistants: A Human-Calibrated Benchmark and Evaluation Audit

First Author | Manuscript under double-blind peer review

- Defined Unwarranted Pragmatic Inference (UPI) as unsupported social assumptions introduced when LLMs write on a user's behalf, and developed a taxonomy separating pragmatic overreach from ordinary task failure.
- Built UPI-Bench with 736 Chinese prompts across ten social writing tasks, and designed a human evaluation of 1,650 anonymized outputs using independent annotation, adjudication, and calibrated automatic judges.
- Found that hidden reasoning reduced human-evaluated UPI by 13.3 percentage points, while judge-guided revision substantially underestimated the same failures, revealing a large gap between automatic and human evaluation.

PhysCrop-Risk: Calibrated Early Low-Yield Alerts from Phenology-Aligned Multispectral Satellite Time Series

First Author | Manuscript under double-blind peer review

- Formulated early crop monitoring as a sequential risk prediction problem with explicit control of false alarms throughout the growing season.
- Developed PhysCrop-Risk, combining Sentinel-2 imagery, weather, and environmental information through crop-specific growing degree day checkpoints, with calibrated alerts and 10 m spatial risk maps.
- Evaluated the system on 2,171 fields across four countries, four crops, and nine seasons; on a 322-field held-out audit, achieved RMSE 1.0582 t ha^{-1}, AUROC 0.8331, and a median 62.5-day warning lead time.

From Predictive Signal to Deployable Value: EHR Measurement-Process Features in ICU AKI Prediction

Co-first Author (listed second) & Corresponding Author | International Journal of Medical Informatics, 221 (2026), 106668

- Led the study design and formal analysis of whether EHR measurement process features improve dynamic ICU AKI prediction, using MIMIC-IV for development and temporal testing and eICU-CRD for external validation.
- Designed robustness and transportability analyses using patient bootstrap, hospital level evaluation, and model sensitivity testing, showing that workflow features added limited predictive value beyond clinical creatinine trajectories.

AlphaMaster: Dual-Chain Feedback for Scalable and Diverse Alpha Factor Discovery

First Author | ACM SIGKDD Conference on Knowledge Discovery and Data Mining

- Proposed AlphaMaster, an automated factor research framework that connects hypothesis generation, backtesting, and iterative refinement to improve the diversity and generalization of quantitative factors.
- Designed the factor discovery system by combining reinforcement learning, GFlowNet, LLM based symbolic generation, and genetic programming, with correlation constraints and multi objective feedback for factor selection and refinement.
- Evaluated AlphaMaster on CSI 500 and CSI 1000 data from 2010 to 2025 under realistic transaction costs, turnover limits, and execution delay, outperforming GP, DSO, AlphaGen, AlphaForge, and LLM based baselines.

INTERNSHIP EXPERIENCE

AI Research & Engineering Intern, Heilongjiang Agricultural Investment Big Data Technology Co., Ltd. **2026.07 – 2026.09**

- Productized PhysCrop-Risk into an operational agricultural risk-warning system, building a Python/SQLite/HTTP backend with standardized evidence ingestion, leakage prevention, hierarchical permissions, asynchronous reporting, and audit logging; the research resulted in a first-authored manuscript under double-blind peer review.
- Developed a DeepSeek-powered constrained reporting agent and enterprise workbench for evidence visualization, report generation, and human review, with factual-consistency and authorization safeguards preventing the LLM from altering predictive evidence or prescribed actions.

Quantitative Research Intern, Financial Quant Group, CCB Fintech **2025.12 – 2026.05**

- Built a multi-source financial data platform covering 30,000+ securities across seven asset classes, with adaptive API probing, automated schema discovery for 1,500+ fields, cross-source field standardization, incremental storage, and optimized bulk ingestion.
- Developed a VeighNa-based multi-channel quantitative trading system with unified CTP/XTP/TQSDK gateways, real-time spread monitoring, automated order execution, market-state detection, and trading-risk controls including permission validation, stop-loss, and auto-close mechanisms.

SELECTED COMPETITIONS & PROJECTS

Kaggle: Hull Tactical - Market Prediction *Team Leader, Top 1.45%, 2026*

Kaggle: DRW-Crypto Market Prediction, Kaggle Community Competition *Team Leader, Top 7%, 2025*

Asia and Pacific Mathematical Contest in Modeling *Team Leader, Second Prize, 2025*

WorldQuant Challenge *Gold & Expert Level; BRAIN Research Consultant, 2025*

SKILLS

Languages: English (IELTS 6.5)
Programming & ML: Python, PyTorch, scikit-learn, LightGBM, XGBoost, SQL
Data & Tools: MySQL, SQLite, HTTP APIs, LaTeX

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